ScienceVision: Can AI Observe and Reason Like a Scientist?

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Abstract

Scientific questions often require interpreting 3D evidence, such as protein contacts or spatial relationships in anatomy. Existing multimodal benchmarks largely evaluate text and 2D images, while scientific 3D evaluations use different representations and protocols across disciplines. These differences complicate cross-disciplinary comparison. To address this gap, we introduce **ScienceVision**, a cross-domain benchmark for scientific 3D understanding and reasoning through expert-level visual question answering (VQA). It contains 250 questions, with 50 each from biology, chemistry, medical imaging, mechanical CAD, and architecture. Each task pairs a native 3D artifact with a scientific question, a specified response format, and a task-specific reference answer, requiring domain knowledge to be applied to geometric or topological evidence. We evaluate 12 models on the same questions under three explicit evidence-access conditions: programmatic analysis of source artifacts, the same access augmented with expert-inspired inspection and measurement tools through the Model Context Protocol (MCP), and fixed six-view renderings as an image-based control. The inspection environment supports view changes, target isolation, and slice inspection. In this main comparison, the best mean answer score is 0.53 out of 1. Source access outperforms fixed views for every model, by 17.2 percentage points on average. Additional MCP tools are associated with smaller score changes for every model, with no consistent direction. Measured performance is thus more sensitive to access to native 3D evidence than to additional inspection tools, making these access conditions essential to interpreting cross-disciplinary 3D understanding and reasoning scores.

HOMEPAGE

1 Introduction

Scientific visual question answering connects visual or structural evidence with domain-specific interpretation (Burgess et al., 2025; Mandiga et al., 2026). In molecular visualization, useful structural information can be distributed across views, with some structures occluded in an individual view (Xiang et al., 2024). One chemistry example in our benchmark asks the solver to identify two atoms from their visibility in the initial view and find the shortest bonded path between them. A medical example asks the solver to locate a specified abnormality and report spatial measurements in the required coordinate frame (Section 2).

Recent evaluations study these requirements in scientific question answering and visualization. DeepTumorVQA organizes questions about 3D CT into recognition, measurement, visual reasoning, and medical reasoning (Chen et al., 2026). ProtStructQA links protein structural measurements and labels to formal task definitions and answer criteria (Mandiga et al., 2026). SciVisAgentBench evaluates scientific analysis and visualization workflows, including analytical questions and spatial outputs (Ai et al., 2026). Appendix A.9 gives the complete comparison.

We introduce **ScienceVision**, a benchmark for scientific 3D understanding and reasoning formulated as visual question answering (VQA). Its questions cover architecture, biology, chemistry, mechanical CAD, and medical imaging. Each task pairs a native 3D object with a question, explicit response requirements, and a reference answer. Submitted answers are scored deterministically using predefined, task-specific matching rules, including accepted equivalents and numerical tolerances where specified. Expert-level describes the domain knowledge and conventions required to interpret the tasks.

Models can use programmatic analysis and interactive inspection to obtain evidence before answering. We evaluate three evidence-access conditions: programmatic source analysis (**Source**), source access augmented with MCP inspection tools (**Source+MCP**) (Model Context Protocol, 2025), and six fixed rendered views (**Fixed6**). Source+MCP retains source access and adds viewer-dependent operations for changing views, isolating targets, navigating slices, and measuring geometry. All three conditions share the VQA questions, response requirements, and deterministic scoring rules.

We compare 12 models under these conditions. Source access yields higher scores than Fixed6 for every model. The score changes associated with adding MCP tools are smaller than the Source–Fixed6 differences for every model and vary in direction. These comparisons concern the evaluated model–harness configurations, which differ in available evidence and permitted computation. Domain and task-family analyses, reference sensitivity checks, and supplementary execution settings further qualify the results.

Our contributions are:

- **A cross-disciplinary benchmark for scientific 3D understanding and reasoning.** *ScienceVision* contains 250 expert-level VQA questions across five equally represented domains and 22 subdomains. Task-specific response specifications, reference answers, and deterministic matching rules make scientific judgments grounded in geometric or topological evidence checkable.
- **An expert-inspired MCP inspection environment.** Viewer-dependent interfaces expose viewpoint control, target isolation, slicing, and scientific measurements. These operations let agents select and inspect evidence needed to answer domain-specific structural questions.
- **A 12-model comparison under explicit evidence-access conditions.** We quantify performance differences between programmatic source analysis, source access with MCP inspection tools, and fixed six-view renderings. Across all models, the difference between source access and fixed views exceeds the absolute score change associated with adding MCP tools, making evidence access a key dimension for interpreting scientific 3D question-answering performance.

2 The ScienceVision Benchmark

2.1 Tasks and scientific evidence

Tasks range from direct structural readings and counts to domain-specific judgments. Each pairs a native 3D object with a scientific question, answer requirements, and a reference answer. The question identifies a target through a scientific description or its appearance in a specified view. The answer requirements define the requested quantities or identities, units, coordinate frame, and conventions. Responses range from a single category to several related measurements or locations; answer entries need not coincide with independently scored components.

Scientific judgment enters through the relation between a target, its structure, and the requested conclusion. In the chemistry example in Figure 1, initial-view occlusion identifies two atoms. Their identities select a shortest bonded path; the solver must return its intermediate atom IDs and the minimum absolute dihedral angle over consecutive four-atom segments. The stored path minimum is 64.49°. Choosing different atoms changes the scientific quantity being measured.

The medical example supplies the diagnostic target, a small left pleural effusion, and requests its extent along the three array axes, all longest axes, and a signed left–right centroid offset. The reference extent

is [86, 81, 61] voxels. Extents include both bounding voxels, whereas the centroid offset is a physical distance in millimetres. These examples illustrate how target identity and scientific conventions determine an answer; they do not define a common process followed by all tasks.



(a) Chemistry: initial view.

(b) Chemistry: labels and measurements.

(c) Medical: CT inspection.

Figure 1: Original recorded inspection states for the chemistry (a,b) and medical imaging (c) examples.

2.2 Coverage

The five equally represented domains contain 22 subdomains, with four to 25 tasks per subdomain. Architecture includes spatial continuity and dimensional judgments; biology includes contacts and spatial relations; chemistry includes conformation and molecular geometry; mechanical CAD includes composition and component properties; medical imaging includes localization, slice comparison, and measurements. Equal domain quotas balance their aggregate weights, not the frequency of scientific questions in practice or intrinsic task difficulty.

Appendix A provides the complete subdomain distribution, answer organization, and five-domain examples. Reference inspections are available for all 250 tasks, with 2,156 recorded operations excluding the initial states. They document expert activity, rather than a minimum reasoning length or a human performance baseline.

2.3 Construction and reference validity

Objects are checked for question support and viewer access. Candidate questions are prepared from templates or curated drafts, sometimes with agent assistance. Experts review targets and domain assumptions, revise questions, inspect the objects, and supply reference answers. Quality review accepts the annotations or returns them for revision. Appendix B.1 preserves the detailed construction account.

The references comprise 110 recorded values, 106 script-derived values, and 34 tool-derived values. These categories describe provenance, not success rates for independently recomputing answers. The scoring configurations contain 54 task-specific adjustments, including component rubrics, aliases, and numerical tolerances. A reference value alone does not specify all accepted alternatives. Appendix B separates construction, reference evidence, and scoring scope.

A prior independent audit found underspecified targets or counting units in eight CAD questions without establishing numerical reference errors. We retain those tasks in the reported benchmark and assess their influence separately in Section 4.3. File consistency, reference provenance, and a partial structural audit support different checks; none certifies the correctness of every reference.

3 Evaluation Design

3.1 Evidence-access conditions

Source supplies the native object and permits code execution on its source data, including geometry and structural calculations. **Source+MCP** adds viewer inspection and measurement tools while preserving source access. MCP use is optional, so the condition permits either computation or visual inspection. **Fixed6** replaces the native object with six pre-rendered views at fixed camera positions and in a fixed order; agents cannot request another view. Medical tasks pair volume renders with slices, yielding twelve images for six nominal views.

These conditions specify different evidence representations and operations, not identical visual inputs. Source–Fixed6 differences therefore do not isolate active observation. Complete access descriptions and the limits of run-specific tool verification appear in Appendix C.

3.2 Models and execution configurations

We evaluate the 12 models in Table 1 on the same 250 tasks under all three conditions. Each configuration combines a model, an agent harness, and evidence access. Codex is assigned to GPT-6 Astra and GPT-6.1 Sol, and Claude Code to the other ten models (OpenAI, 2026; Anthropic, 2026). GPT Codex uses maximum reasoning effort; Claude Code effort is not recorded. Study-level Fixed6 harness assignments cannot be independently verified for every retained run.

The protocol specifies three repeats per task. Gemini has some missing repeats under both source conditions, and Hunyuan has three tasks with two repeats under Fixed6. Earlier outputs were retained alongside declared reruns of 39 tasks after question or scoring changes. These records constrain comparisons across inputs, scorers, budgets, and model versions. Appendix D gives the exact coverage and available execution evidence; supplementary alternative-harness runs remain separate from the canonical leaderboard.

3.3 Scoring and aggregation

For task i and repeat r, task-specific rules grade the submitted answer against its reference, yielding

$$s_{ir} \in [0, 1]. \tag{1}$$

The rules specify scored components, matching, and partial credit. They need not average all stored answer entries, and we do not apply a new universal eligibility gate to the exported scores. Appendix B documents this scope.

Let R_i contain the available scored repeats. We first average within each task, then weight tasks equally:

$$\bar{s}_i = \frac{1}{|R_i|} \sum_{r \in R_i} s_{ir}, \qquad S_D = \frac{1}{|D|} \sum_{i \in D} \bar{s}_i. \tag{2}$$

Here $|D| = 50$ for a domain and 250 for the full benchmark. Missing repeats are omitted, recorded zeros are retained, and intermediate values are not rounded. Mean task scores retain partial credit and differ from complete-answer accuracy.

Cross-configuration score differences are descriptive comparisons of the evaluated settings. They do not establish a controlled MCP effect. Appendix E provides complete scores and Appendix D documents the available execution records.

4 Empirical Findings

4.1 Performance under three evidence-access conditions

Table 1 reports the 36 canonical model–access scores. GPT-6 Astra/Codex has the highest Source score, 0.5291, while Claude Fable 5.1 has the highest Fixed6 score, 0.2965. Source exceeds Fixed6 for every

model by 7.73–28.49 percentage points; Source+MCP also exceeds Fixed6 in every row. These are differences between evaluated configurations, including permitted source computation, rather than isolated benefits of visual inspection.

Table 1: Mean task scores for 12 models under three access settings on the same 250 tasks. Available repeats are averaged within each task, then tasks are weighted equally. Scores are displayed to four decimals. Supplementary runs with alternative harnesses are reported separately.

Model	Source	Source+MCP	Fixed6
GPT-6 Astra	0.5291	0.5282	0.2498
GPT-6.1 Sol	0.4893	0.5110	0.2044
Claude Fable 5.1	0.4619	0.4631	0.2965
Claude Opus 5.5	0.4539	0.4343	0.2383
Gemini 4 Argon	0.4278	0.4321	0.1462
Qwen3.8 Max 0902	0.3830	0.3736	0.2450
Qwen3.8 Flash	0.3616	0.3681	0.2468
Kimi K3	0.3609	0.3369	0.2148
GLM-5.3 Flash	0.2846	0.2858	0.1521
DeepSeek V4 Flash	0.2652	0.2678	0.1661
Doubao Seed Evolving	0.2435	0.2527	0.1102
Hunyuan-4 Preview	0.1598	0.1738	0.0824

The strongest observed answer score in the main comparison remains about one half of the available credit. This score allows partial answers, so it does not imply that half the questions were answered completely. Appendix E.1 retains configuration details and both source-access and Fixed6 leaderboard displays; they are alternative presentations of the same scores.

4.2 MCP score differences and task heterogeneity

MCP changes are smaller than Source–Fixed6 differences and have no common direction. Eight configurations increase and four decrease, ranging from -2.40 percentage points for Kimi K3 to $+2.17$ points for GPT-6.1 Sol. Astra changes by -0.08 points.

Comparing Source with Source+MCP, biology scores rise for 11 of the 12 canonical configurations, with an equal-model mean difference of $+3.26$ percentage points; chemistry scores fall for nine, with a mean difference of -1.91 points. Domain changes can also cancel within a model: Astra's biology score rises from 0.6200 to 0.6778, while medical imaging falls from 0.3783 to 0.3406. Sol's biology score rises from 0.6000 to 0.6667. Among the canonical configurations, Fable 5.1 with MCP reaches the highest domain mean, 0.6989 in biology. Figures 6 and 7 retain all 180 domain means in the appendix.

Task-family analysis sharpens this contrast. Equal-weight averages over the 12 canonical models show Source–Fixed6 gaps of 45.06 percentage points for dihedral measurement ($N = 9$) and 40.68 points for conformation/occlusion ($N = 21$). Every model scores higher with Source in both families, connecting the aggregate result to the molecular-geometry judgments in Section 2. These differences remain descriptive comparisons of available evidence and computation.

On 25 lesion-pointing tasks, corresponding means are 0.1011, 0.0967, and 0.0344 for Source, Source+MCP, and Fixed6. All 36 configurations score lower there than on slice comparison. Each model also has positive and negative MCP differences across the 22 subdomains. Thus, a single domain or overall mean does not describe all scientific judgments. Appendix E.3 gives the aggregation definitions and all 792 subdomain values; these small, selected families are not population-level ability estimates.

4.3 Reliability and interpretation of the findings

Excluding the eight CAD questions with target or counting-unit ambiguity leaves 242 tasks. All three model rankings, every Source–Fixed6 direction, and the eight-increase/four-decrease MCP split remain unchanged. Astra Source changes from 0.5291 to 0.5356, and Fable Fixed6 from 0.2965 to 0.3063. This

sensitivity check preserves the original benchmark and does not validate all remaining references; Appendix E gives the complete diagnostic.

Execution settings also affect the apparent score ceiling. Supplementary Opus/Codex Source (0.5448) and Fable/Codex Source+MCP (0.5349) exceed the canonical maxima. Codex scores are higher in the eight available full-set harness comparisons by 4.56–13.40 percentage points, but execution settings and evaluation records differ. They do not establish a controlled harness effect or a universal ordering. Table 13 gives the full comparison; Appendix D.3 gives the recovery case.

More fixed views do not consistently improve scores in the earlier camera study: neither plan gains from six to twelve views (Table 14). This historical pattern does not establish an optimal view budget for the current benchmark.

MCP-only inspection and Source+MCP test distinct access protocols (Table 15). The historical tool-only record does not estimate the marginal benefit of adding MCP to source computation in the current evaluation.

Answer scores cannot identify whether a failure arose from target selection, evidence acquisition, domain inference, or output interpretation. Recorded expert inspections show an intended route rather than an evaluated model's observations. Complete task-score profiles and available trajectory evidence are retained in Appendix E and Appendix D.3; they do not supply a human baseline or a validated process-error taxonomy.

5 Discussion and Conclusion

ScienceVision evaluates scientific judgments about supplied 3D objects. It does not assess the ability to formulate research questions, design experiments, or validate new scientific claims. Its results address the title's broad question within this narrower scope.

Equal domain quotas give each domain the same weight, but do not reflect the frequency of questions in scientific practice. Subdomains and object families remain uneven. Task selection has used prior model performance, so the scores cannot estimate difficulty for randomly sampled scientific questions. Reference inspections and annotation effort also do not establish a human performance baseline.

Reference quality and accepted-answer rules affect how scores should be interpreted. Recorded values, computed references, and tool-derived measurements provide different forms of evidence; their presence alone does not guarantee correctness. Differences in execution settings, incomplete repeats, and uncertainty about earlier runs further limit attribution to any single capability.

The evaluation grades answers rather than observation or reasoning processes. Without component-level verdicts and model trajectories, an incorrect answer cannot be reliably assigned to failed evidence acquisition or incorrect domain inference. Connecting validated process annotations to these trajectories would support that distinction. Providing MCP tools and reporting aggregate scores does not establish that the models used those tools successfully.

ScienceVision evaluates scientific questions about native 3D objects across five domains, using explicit answer requirements and task-specific references. Source access outperforms fixed views for every tested model, while adding MCP tools produces smaller changes in both directions. These comparisons show why evidence access and execution settings matter when interpreting benchmark scores. They do not independently establish correct observation or reasoning processes, or a causal benefit from MCP.

References

Kuangshi Ai, Haichao Miao, Kaiyuan Tang, Nathaniel Gorski, Jianxin Sun, Guoxi Liu, Helgi I. Ingolfsson, David Lenz, Hanqi Guo, Hongfeng Yu, Teja Leburu, Michael Molash, Bei Wang, Tom Peterka, Chaoli Wang, and Shusen Liu. SciVisAgentBench: A benchmark for evaluating scientific data analysis and visualization agents, 2026. URL https://arxiv.org/abs/2603.29139v4.

Anthropic. Claude code documentation, 2026. URL https://code.claude.com/docs/en/overview.

James Burgess, Jeffrey J. Nirschl, Laura Bravo-Sánchez, Alejandro Lozano, Sanket Rajan Gupte, Jesus G. Galaz-Montoya, Yuhui Zhang, Yuchang Su, Disha Bhowmik, Zachary Coman, Sarina M. Hasan, Alexandra Johannesson, William D. Leineweber, Malvika G. Nair, Ridhi Yarlagadda, Connor Zuraski, Wah Chiu, Sarah Cohen, Jan N. Hansen, Manuel D. Leonetti, Chad Liu, Emma Lundberg, and Serena Yeung-Levy. MicroVQA: A multimodal reasoning benchmark for microscopy-based scientific research. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 19552–19564, 2025. URL https://openaccess.thecvf.com/content/CVPR2025/html/Burgess_MicroVQA_A_Multimodal_Reasoning_Benchmark_for_Microscopy-Based_Scientific_Research_CVPR_2025_paper.html.

Yixiong Chen, Wenjie Xiao, Pedro R. A. S. Bassi, Boyan Wang, Liang He, Xinze Zhou, Sezgin Er, Ibrahim Ethem Hamamci, Zongwei Zhou, and Alan Yuille. DeepTumorVQA: A hierarchical 3D CT benchmark for stage-wise evaluation of medical VLMs and tool-augmented agents, 2026. URL https://arxiv.org/abs/2605.09679.

Ziru Chen, Shijie Chen, Yuting Ning, Qianheng Zhang, Boshi Wang, Botao Yu, Yifei Li, Zeyi Liao, Chen Wei, Zitong Lu, Vishal Dey, Mingyi Xue, Frazier N. Baker, Benjamin Burns, Daniel Adu-Ampratwum, Xuhui Huang, Xia Ning, Song Gao, Yu Su, and Huan Sun. ScienceAgentBench: Toward rigorous assessment of language agents for data-driven scientific discovery. *arXiv preprint arXiv:2410.05080*, 2024. URL https://arxiv.org/abs/2410.05080.

egtai. GMX-VMD-MCP: MCP service for GROMACS and VMD molecular dynamics simulations and visualization. Software repository, 2025. URL https://github.com/egtai/gmx-vmd-mcp. Accessed 2026-09-23.

Lawrence Livermore National Laboratory. BioImage-Agent. Software repository, 2025. URL https://github.com/LLNL/bioimage-agent. Accessed 2026-09-23.

Shusen Liu, Haichao Miao, Zhimin Li, Matthew Olson, Valerio Pascucci, and Peer-Timo Bremer. AVA: Towards autonomous visualization agents through visual perception-driven decision-making. *Computer Graphics Forum*, 43(3):e15093, 2024. doi: 10.1111/cgf.15093. URL https://doi.org/10.1111/cgf.15093.

Shusen Liu, Haichao Miao, and Peer-Timo Bremer. ParaView-MCP: An autonomous visualization agent with direct tool use, 2025. URL https://arxiv.org/abs/2505.07064.

Aravind Mandiga, Guoming Li, Jin Lu, Ismailcem Budak Arpinar, Khaled Rasheed, and Samuel E. Aggrey. ProtStructQA: A denotation threshold in protein structural reasoning, 2026. URL https://arxiv.org/abs/2606.00451.

Giorgio Mariani, Alessandro Raganato, Simone Melzi, and Gabriella Pasi. GLUE3D: General language understanding evaluation for 3d point clouds. *Information Fusion*, 129:104007, 2026. doi: 10.1016/j.inffus.2025.104007.

Model Context Protocol. Model context protocol specification, 2025. URL https://modelcontextprotocol.io/specification/2025-06-18.

OpenAI. Codex CLI, 2026. URL https://developers.openai.com/codex/cli.

Tom Peterka, Tanwi Mallick, Orcun Yildiz, David Lenz, Cory Quammen, and Berk Geveci. ChatVis: Large language model agent for generating scientific visualizations, 2025. URL https://arxiv.org/abs/2507.23096.

Mingjia Wang, Taiting Lu, Ziwei Dong, Sisong Bei, Jingying Zeng, Runze Liu, Kaiyuan Lin, Hongxing Pan, Kai Zhang, Yizheng Hou, Yangshoudu Zheng, Chenchen Guo, Weiyuan Meng, Shubin Lyu, Zhijun Zheng, Dexu Wang, Xinyu Bai, Shurui Qian, Zhangzixin, Mengyu Pan, Guoliang Shi, Ling Ma, Yifan Yang, Qi He, Yi-Chao Chen, Yincheng Jin, Sung-Liang Chen, and Mahanth Gowda. OmniCAD: A large-scale benchmark for 3D spatial reasoning in robotics assemblies, 2026. URL https://arxiv.org/abs/2608.22637.

Hongxin Xiang, Li Zeng, Linlin Hou, Kenli Li, Zhimin Fu, Yunguang Qiu, Ruth Nussinov, Jianying Hu, Michal Rosen-Zvi, Xiangxiang Zeng, and Feixiong Cheng. A molecular video-derived foundation model for scientific drug discovery. *Nature Communications*, 15:9696, 2024. doi: 10.1038/s41467-024-53742-z. URL https://www.nature.com/articles/s41467-024-53742-z.

zhaoyouj. MCP-Slicer: 3D Slicer model context protocol integration. Software repository, 2025. URL https://github.com/zhaoyouj/mcp-slicer. Accessed 2026-09-23.

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A Task Specifications, Coverage, and Context

A.1 Subdomain inventory

Table 2 lists the 22 subdomains and their task counts. Each task belongs to one subdomain, although a question may require several kinds of observation or judgment. For example, a conformation question may also ask about an occluded atom. These groups describe task content rather than mutually exclusive scientific abilities. Supporting metadata records the assignment of each task.

A.2 Answer structure and conventions

The reference answers contain 143 single-entry responses and 107 responses with multiple recorded entries. There are respectively 143, 35, 45, 11, 5, 8, 2, and 1 references with 1, 2, 3, 4, 5, 6, 7, and 9 entries. A list counts as one entry, as does a single-word response. These counts describe answer organization and need not match the number of independently requested judgments.

Tasks can request an atom identity, an ordered list, a category, a coordinate, or a measurement. The question defines the convention. Voxel extents need not use physical distance units, and a list of intermediate atoms may exclude its endpoints. The examples below preserve the recorded reference values, without defining all tolerances or equivalent accepted answers.

Table 2: Complete subdomain composition. Objectives are illustrative descriptions of question content; they are not additional per-task ability annotations.

Subdomain	Tasks	Illustrative objective
Architecture (50)		
Dimensional compliance	16	Check dimensions against a stated requirement.
Alignment and stacking	12	Identify continuity or alignment of members.
Interference and clearance	12	Distinguish spatial interference from clearance.
Object geometry readings	10	Read object dimensions, arrangement, or scene geometry.
Biology (50)		
Molecular contact	14	Identify atom contacts and associated distances.
Spatial arrangement	14	Describe the relative arrangement of components.
Interaction typing	8	Identify interaction classes or coordinating atoms.
Proximity choice	8	Select the closest specified component or atom.
Occluded target geometry	6	Locate a view-defined target and assess its geometry.
Chemistry (50)		
Conformation and occlusion	21	Combine molecular conformation with occluded targets.
Dihedral measurement	9	Measure a specified torsion or its minimum value.
Molecular extent	9	Determine the separation of farthest heavy atoms.
Hybridization state	7	Classify and count heavy-atom hybridization states.
Crystal periodicity	4	Analyze a unit cell and its periodic contents.
Mechanical CAD (50)		
Part-instance counting	20	Count component copies, including hidden instances.
Part-property identification	14	Determine a part's geometric properties.
Assembly-composition counting	10	Count repeated parts or designs within an assembly.
Mechanism motion	6	Analyze gears, mechanisms, or assembly relations.
Medical imaging (50)		
Lesion pointing	25	Locate a specified abnormality in a CT volume.
Slice comparison	12	Compare visible anatomy across specified slices.
Geometry measurement	8	Measure a structure's extent or position.
Anatomy pointing	5	Locate a named anatomical structure in a volume.
Total	**250**	**22 subdomains**

A.3 Reference inspection operations

All 250 tasks have a recorded expert inspection, comprising 2,156 tool operations in total. We count viewer operations and exclude the initial state. The ranges 1–2, 3–5, 6–10, 11–20, 21–40, and 41 or more operations contain 36, 95, 62, 32, 21, and 4 tasks, respectively. These counts describe recorded expert activity, rather than model tool use or a minimum reasoning length.

A.4 Chemistry example

The molecular-conformation question first asks for the smallest absolute dihedral angle formed by four consecutively bonded atoms with element sequence H-C-O-C. It then identifies two atoms by their occlusion in the initial view. The solver must report those atoms, the ordered intermediate atoms on their shortest bonded path, and the number of intermediate atoms. If the path has at least four atoms, the solver also reports its smallest absolute dihedral angle over consecutive four-atom segments. Otherwise, no path angle is reported.

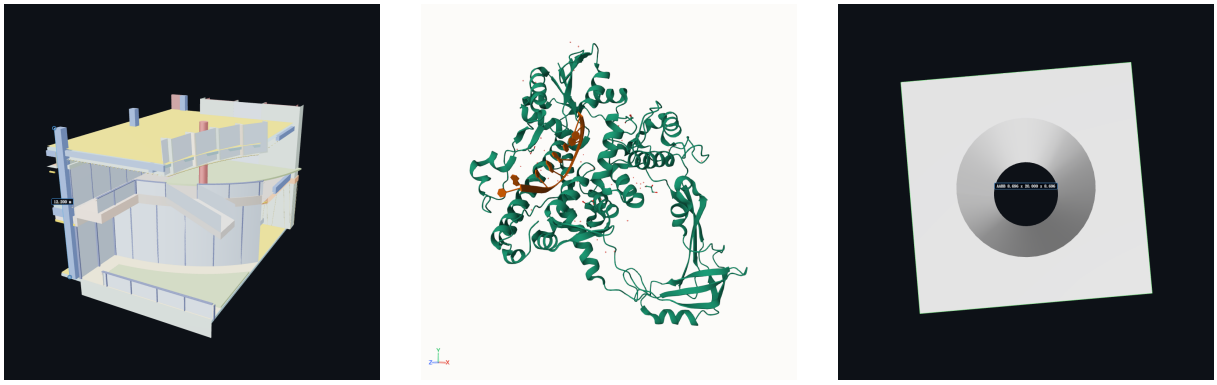
The reference gives 53.34° for the minimum H-C-O-C angle. The nearly fully occluded atom is 21, and the partially occluded heavy atom is 2. Their path contains three intermediate atoms, ordered 8, 9, and 3, with a minimum absolute path dihedral angle of 64.49°. These two angles answer different parts of the question. Figure 1 shows the initial view in panel (a) and the later labeled inspection in panel (b).

A.5 Medical example

The CT question identifies a small left pleural effusion and asks the solver to locate its centroid and representative slice. It also requests the target's extent along the three sampling-grid axes, every axis with the greatest extent, and the signed left–right centroid offset. Extents include both bounding voxels, and tied longest axes must be listed in increasing order. Measurements use at most six decimal places.

The recorded reference gives extents of $[86, 81, 61]$ voxels, identifies axis 0 as the longest, and gives a signed offset of -64.56 millimetres. Axis indices identify the sampling grid, rather than anatomical directions. Voxel extents should not be interpreted as physical lengths without the corresponding coordinate definition.

A.6 Examples from architecture, biology, and mechanical CAD



(a) Architecture. **(b)** Biology. **(c)** Mechanical CAD.

Figure 2: Recorded reference inspections for examples from architecture, biology, and mechanical CAD.

Architecture: continuity across storeys. The question asks which vertical structural members extend continuously across at least two storeys. It requests their count, shortest and longest vertical extents, and whether discontinuous members exist. Coordinates use millimetres with the vertical direction as the third axis. The viewer instead uses metres with the vertical direction as its second axis. Converting between them swaps the second and third coordinates, negates the new second coordinate, and multiplies all coordinates by 1,000.

The reference identifies four continuous members, with shortest and longest extents of 12,000 and 12,200 millimetres, respectively. It reports no discontinuous members.

Biology: terminal residues and chain position. The question asks whether the N-terminus or C-terminus of the largest polymer chain is closer to its centre. Chain size is measured by atom count, with ties resolved by the earliest chain identifier. The centre is the alpha carbon of the middle residue, choosing the later middle residue when the chain length is even. Terminal positions use the alpha carbons of the first and last residues in file order. The task uses the first structural model and the highest-occupancy alternate positions, excluding hydrogens and solvent. The reference answer is C-terminus.

Mechanical CAD: a component's geometry. The question asks whether a component is essentially planar or bends substantially out of any single plane. It requests a single category, planar or non-planar. The reference answer is non-planar, illustrating how one recorded answer entry can correspond to a single-word response.

A.7 Coverage and recorded inspection distribution

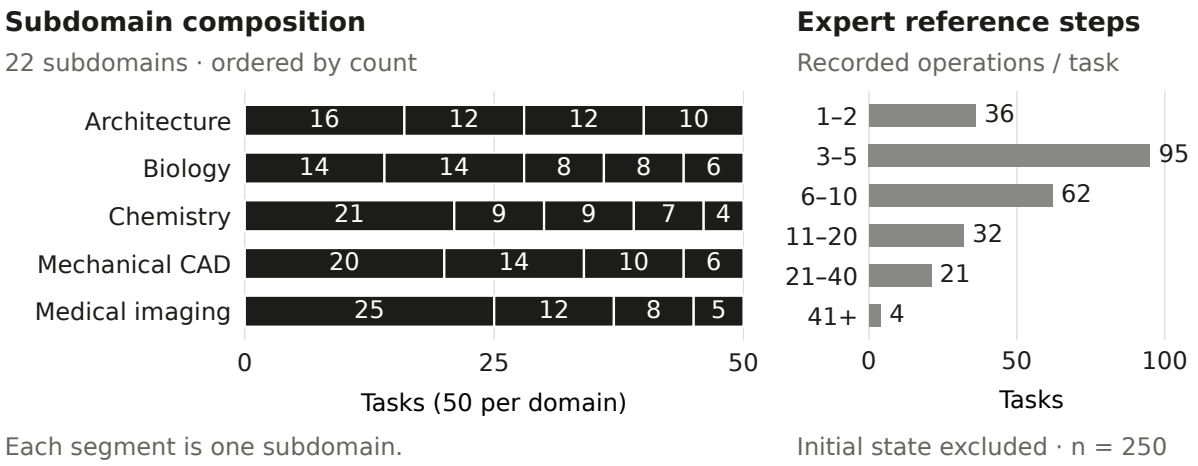


Figure 3: Subdomain composition (left; names in Appendix A) and expert reference inspection lengths (right).

A.8 Extended task-design description

ScienceVision asks for structural and spatial judgments about specific 3D scientific objects. Its 250 tasks are evenly distributed across architecture, biology, chemistry, mechanical CAD, and medical imaging (Figure 3). The questions concern properties of particular objects, such as continuity across storeys in a building, contact between macromolecular structures, or the shape and multiplicity of assembly components.

A.8.1 Objects, targets, and answer requirements

Each task pairs an object with a question, answer requirements, and a reference answer. The question can identify its target by a scientific description or by its appearance in a specified view. The requirements specify what to report and define the relevant units, coordinates, and object-identity conventions. Responses range from a single category to several related quantities or locations. Expert inspection records preserve the viewer operations and visual states associated with the reference annotations; Appendix A details both the records and answer formats.

A.8.2 From target selection to scientific judgment

Chemistry: a view-defined target and its bonded geometry. In the chemistry example (Figure 1a,b), the initial camera view defines two atoms by their occlusion. Their identities select the endpoints of a shortest bonded path. The task requests the intermediate atom IDs, excluding both endpoints, and the smallest absolute dihedral angle over consecutive four-atom segments along the path. The stored reference gives 64.49° for this path minimum.

Medical imaging: a named lesion and its spatial extent. The medical example supplies the diagnostic target, a small left pleural effusion in a CT volume. It asks for the lesion's extent along the three array axes, every longest axis, and a signed left–right centroid offset. The reference extent is $[86, 81, 61]$ voxels. Extents include both bounding voxels, whereas the centroid offset is a physical distance in millimetres.

A.9 Scientific question answering and evidence-access benchmarks

A.9.1 Scientific structural question answering

DeepTumorVQA organizes questions about 3D CT into recognition, measurement, visual reasoning, and medical reasoning, with reference answers generated by deterministic programs from validated imaging metadata (Chen et al., 2026). It reports multiple-choice answer accuracy and rule-based scoring of free-form responses. ProtStructQA defines questions over AlphaFold-predicted proteins using formal

task definitions, covering structural measurements and labels (Mandiga et al., 2026). Deterministic calculations supply each reference value, and predictions are checked using answer-type-specific tolerances, exact matching, or set overlap. Both benchmarks connect object-derived evidence to explicit answer criteria within a scientific domain. *ScienceVision* uses task-specific references and answer requirements across five scientific and engineering domains, specifying requested quantities, identities, and locations under the relevant domain conventions. Table 3 compares scientific scope, question-answering objectives, evidence access, and formal scoring across related benchmarks.

Table 3: Comparison of benchmark scope, question-answering (QA), evidence access, and grading. ✓: reported; ×: not reported. Cross-domain denotes multiple scientific fields. QA focus includes captioning classified as QA by the benchmark authors; Partial denotes QA alongside visualization or processing tasks. Interaction includes agent-selected views, slices, crops, or rendered prediction feedback. Static-image QA denotes a separate fixed-image evaluation. Deterministic scoring considers formal outcomes: ✓, rules without LLM judges; Partial, mixed rule-based/LLM grading; ×, LLM-only grading.

Benchmark	Cross-domain science	QA focus	Visual interaction	Static-image QA	Deterministic scoring
DeepTumorVQA (Chen et al., 2026)	×	✓	✓	✓	✓
ProtStructQA (Mandiga et al., 2026)	×	✓	×	×	✓
OmniCAD (Wang et al., 2026)	×	×	✓	×	✓
SciVisAgentBench (Ai et al., 2026)	✓	Partial	✓	×	Partial
GLUE3D (Mariani et al., 2026)	×	✓	×	✓	Partial
***ScienceVision* (ours)**	✓	✓	✓	✓	✓

A.9.2 Scientific visualization agents and evidence access

SciVisAgentBench evaluates scientific analysis and visualization workflows in established software, including analytical question answering and spatial outputs (Ai et al., 2026). It grades produced outcomes against expert-authored rubrics using multimodal judges and deterministic evaluators. *ScienceVision* centers its task families on scientific questions about supplied artifacts and grades submitted answers against task-specific references using deterministic checks without an LLM judge. The distinction concerns the task objectives and graded outcomes; question answering and interaction are already present in prior scientific 3D evaluations.

Scientific agents obtain evidence through generated analysis scripts (Chen et al., 2024; Peterka et al., 2025), visual feedback, and direct viewer operations (Liu et al., 2024; 2025). MCP interfaces also expose operations for molecular visualization, bioimages, and medical volumes (egtai, 2025; Lawrence Livermore National Laboratory, 2025; zhaoyouj, 2025). *ScienceVision* uses these access mechanisms as evaluation conditions: source data with code execution, the same access with additional MCP inspection tools, and fixed rendered views. These conditions distinguish the evidence a solver receives and the operations it may request from the final answer being graded. Tool availability alone does not establish how a particular answer was obtained.

A.10 Extended motivation and resource positioning

Scientific observation begins with a question about an object and the evidence needed to answer it. Identifying atoms hidden in an initial molecular view can determine which bonded path must be analyzed. Locating a lesion in a CT volume requires interpreting anatomy and returning a location in the specified coordinate frame. The task is to connect evidence from the supplied object to a domain-specific conclusion.

For an AI system, this connection can involve selecting a target, revealing a structure, measuring a relation, and applying scientific definitions or conventions. The relevance of a measurement depends on which object was identified and what the question asks. A correct distance between the wrong atoms does not answer a molecular-contact question. Likewise, a plausible anatomical description does not supply a requested voxel location.

Existing evaluations already study scientific structural question answering and agent interaction with native 3D data. DeepTumorVQA evaluates questions about CT volumes, and SciVisAgentBench includes scientific analysis, visualization, and question-answering outcomes across domains (Chen et al., 2026; Ai et al., 2026). ScienceVision evaluates scientific questions about supplied 3D objects using checkable answers. It brings five domains into a shared evaluation, allowing comparison across evidence-access conditions while documenting execution differences.

ScienceVision contains 250 questions across architecture, biology, chemistry, mechanical CAD, and medical imaging, with 50 questions per domain and 22 subdomains. The questions combine structural evidence with domain concepts, object identity, or explicit conventions. Expert-level describes the intended knowledge required to interpret a task; it does not denote measured difficulty relative to a human baseline.

ScienceVision evaluates scientific visual question answering (VQA), with predefined questions, response requirements, and reference answers. Models can acquire evidence through programmatic source analysis, source access augmented with MCP inspection tools (Model Context Protocol, 2025), or six fixed rendered views. Code execution and viewer operations support the answering process, while submitted answers are scored deterministically against task-specific references using predefined matching rules.

Our contributions are:

- **A cross-disciplinary benchmark for scientific 3D understanding and reasoning.** *ScienceVision* contains 250 expert-level VQA questions across five equally represented domains and 22 subdomains. Task-specific response specifications and reference answers make scientific judgments grounded in geometric or topological evidence checkable.
- **An expert-inspired MCP inspection environment.** Viewer-dependent interfaces expose viewpoint control, target isolation, slicing, and scientific measurements. These operations let agents select and inspect evidence needed to answer domain-specific structural questions.
- **A 12-model comparison under explicit evidence-access conditions.** We quantify performance differences between programmatic source analysis, source access with MCP inspection tools, and fixed six-view renderings. Across all models, the difference between source access and fixed views exceeds the absolute score change associated with adding MCP tools, making evidence access a key dimension for interpreting scientific 3D question-answering performance.

B Construction, References, and Scoring

B.1 Construction and reference evidence

B.1.1 Question construction and expert annotation

We first check that each 3D object supports the intended question and can be inspected in the viewer. Candidate questions come from templates or curated drafts, sometimes with agent assistance. Questions that satisfy the task rules are selected and sent to experts with their supporting objects.

Experts review the targets and domain assumptions, revise the questions where needed, and inspect the 3D evidence to provide reference answers. Inspection records document how they obtained those answers. A subsequent quality review accepts the annotations or returns them for revision.

B.1.2 Reference answers and recorded inspections

Each task includes a reference answer, while inspection records show how experts examined the object. Some references are derived through computation or measurement tools; others are recorded directly. These resources support answer checking, but a recorded reference alone does not establish which alternative answers should receive credit.

Of the 250 references, 110 are recorded values, 106 are derived by scripts, and 34 are derived through tools. These categories describe how the references were obtained. They do not establish that every answer can be recomputed from the supplied objects, because not all reference calculations and scoring procedures were re-executed.

Inspection records describe one route to an answer, rather than the minimum work required or the route an agent must follow. Recorded viewer actions also differ from model responses and tool calls. They do not provide a human task-success baseline.

B.1.3 Verification and accepted answers

Scoring must account for units, coordinates, and equivalent representations of an answer. A lesion-pointing question may admit several locations, an object may have accepted aliases, and a numerical answer may require a task-specific tolerance. A single reference value does not define all these rules.

The task configurations contain 54 task-specific scoring adjustments, including component rubrics, accepted aliases, and numerical tolerances. These categories overlap. The configurations document intended scoring rules, but do not verify every scoring decision in earlier runs. Section 3 defines the scoring unit and aggregation.

B.2 Task-specific scoring and verification scope

Task scores follow Equation 1. Scoring rules specify how answers are interpreted, which equivalent labels are accepted, and how components are matched and weighted. Some tasks score only a subset of the recorded answer components. The available documentation does not establish a universal averaging rule or a common rule for excluding responses from scoring. We retain the recorded scores without applying an additional exclusion rule.

C Access Interfaces and Fixed Views

The inspection interface supports camera control, target selection and isolation, clipping or slice navigation, and scientific measurements. Available operations depend on the viewer and may differ between the expert and agent interfaces. Lists of permitted operations do not verify which tools were available or functioning in every run. Source+MCP also permits computation from source data, so its score does not show whether visual inspection supplied the decisive evidence.

C.1 Evidence-access conditions

Source supplies the native 3D object and permits computation from its source data, including distances and structural relations. **Source+MCP** adds viewer inspection and measurement tools. Tool use is optional, so this condition evaluates a combined workflow that can use either source analysis or visual inspection.

Fixed6 supplies the question and six pre-rendered views in place of the native object. The camera positions and image order are fixed; agents cannot request another view. Medical tasks pair volume renders with slices, giving twelve images for six nominal views. Appendix C describes the interfaces and documentation limits. An earlier comparison of aligned and orbit views is analyzed separately because it

used a different task set.

D Configurations and Repeat Coverage

All main results use the same recorded scores, covering 9,000 task–configuration pairs: 250 tasks for each of 12 models under three access conditions. We check task membership, domain assignments, repeat counts, and agreement between task means and their available repeats. These checks reproduce all 36 reported overall means. Missing repeats are omitted, and intermediate values retain full precision.

The blog displays the same 36 overall means but rounds task means before computing domain scores. This produces one last-digit difference: Astra Source in chemistry is 0.6094 here and 0.6095 on the blog. The difference comes from rounding rather than a new evaluation. Reproducing recorded scores does not independently validate the original runs or scoring decisions.

D.1 Execution settings and incomplete records

The study assigns Codex to GPT-6 Astra and GPT-6.1 Sol, and Claude Code to the other ten models under all three conditions. GPT Codex uses maximum reasoning effort; Claude Code effort is not recorded. Fixed6 records do not identify the harness for every run, so the study-level assignment cannot be independently verified throughout.

Earlier outputs were retained alongside reruns of 39 tasks after question or scoring revisions. The present task set includes 205 original tasks and 45 replacements. Shared task identifiers do not establish that questions, references, and inputs remained identical. The available records cannot fully reconstruct the conditions for every retained run.

There are 26,653 available repeat scores. For Gemini under each source-access condition, 172 tasks have two repeats and 78 have three. For Hunyuan Fixed6, three tasks have two repeats and 247 have three. Every other configuration has three scores per task. These counts establish score availability, rather than independence between repeats.

The supplementary Fable/Opus Codex runs use high reasoning effort because maximum-effort requests encountered timeouts. Their reasoning budgets therefore differ from those of the GPT Codex runs. We report model responses and completion tokens separately; hidden reasoning-token counts are unavailable.

D.2 Data for the harness comparison

Table 13 combines the main GPT/Codex and Claude/Claude Code results with supplementary runs using the alternative harness. The GPT/Claude Code records precede the final 39-task reruns. The Claude/Codex records contain 3,000 trials spanning two models, two access conditions, and three repeats on 250 tasks.

All 16 table cells cover the same 250 task identifiers with three scores per task. We average repeats within each task, then weight tasks equally and retain zero scores. Shared identifiers do not guarantee identical questions, references, inputs, or scoring conditions. High effort is documented for all supplementary Claude/Codex trials; effort remains unknown for the main Claude Code runs.

Proposed revised GPT/Claude Code averages could not be reproduced from the available trial-level corrections. We therefore use the complete per-task records. All displayed means and percentage-point differences are calculated before rounding, without replaying the original runs.

D.3 Interaction length, tool feedback, and timeouts

All 3,000 supplementary Fable/Opus Codex trials include readable trajectories and completion token counts. High reasoning effort is recorded throughout. We count each recorded model message as one response, including the final answer but excluding initialization messages. Tool calls are counted within those responses. Completion tokens measure generated output; the available records do not measure hidden reasoning length.

The 1,243 full-score trials have median values of six model responses and 2,419 completion tokens. The 1,061 zero-score trials have medians of nine responses and 4,628 tokens. The remaining 696 trials receive partial credit. These are descriptive associations across tasks, models, and conditions, rather than evidence that shorter interactions improve answers. There are 47 execution timeouts, each at a 3,600-second limit. These differ from the request timeouts that motivated the high-effort setting. An earlier reasoning-budget pilot lacks recoverable pairing and denominators, so we do not use its reported rates.

One Fable/Codex Source trial illustrates recovery from failed data access. After a molecular-analysis library could not be loaded, the agent inspected the atom records and read the data by an alternative method. It selected the largest occupied component excluding water and polymers, then computed the closest atom-pair distance. The answer, 2.229 angstroms, received full credit under the archived rubric. The sequence contains six model responses and five tool calls. It shows how feedback can support recovery, without establishing a causal benefit from longer interaction or a difference from Claude Code.

The selected pair joins magnesium and histidine. It was not excluded as bonded because the molecular file contained no explicit connectivity records. This reproduces the recorded calculation and score, but does not validate the intended definition of a nonbonded contact.

Codex records combine inspection and computation through general execution tools. The available Claude Code records expose separate reading, editing, and execution operations. This difference concerns how tools are organized. The main Claude Code score records lack matching trajectories, preventing attribution of score differences to operation order or granularity.

D.4 Models and execution configurations

We evaluate the 12 models in Table 1 on 250 tasks under all three access conditions, with 50 tasks in each domain. Each configuration combines a model, an agent harness, and an access condition. The harness manages file access, computation, and tool use.

We use Codex for GPT-6 Astra and GPT-6.1 Sol, and Claude Code for the other ten models (OpenAI, 2026; Anthropic, 2026). The GPT models use maximum reasoning effort; Claude Code effort is not recorded. For Fixed6, harness assignments are documented for the study but cannot be verified for every run. Appendix D gives the available configuration records.

The protocol specifies three repeats per task. Some repeats are missing for Gemini in both source conditions and for Hunyuan in Fixed6. Appendix D gives the counts. Supplementary runs with alternative harnesses are analyzed separately in Section F.1.

E Complete Results and Sensitivity Analyses

E.1 Main scores and execution settings

The 36 canonical scores are reported once in Table 1 in the main text.

E.1.1 Additional context for the canonical scores

Figure 4 compares the 12 selected model–harness configurations under Source and Source+MCP. Appendix E.1 gives all three settings and their configurations. Among these canonical leaderboard rows, GPT-6 Astra/Codex has the highest Source score, 0.5291, while Claude Fable 5.1 has the highest Fixed6 score, 0.2965. Source exceeds Fixed6 for every model, by 7.73–28.49 percentage points (Table 1).

These mean task scores include partial credit. Source access permits direct calculation, and execution records are incomplete, so the gaps do not isolate the benefit of visual inspection. Rankings also depend on the execution setting. Supplementary Claude Opus 5.5/Codex Source (0.5448) and Claude Fable 5.1/Codex Source+MCP (0.5349) scores exceed the main table's maxima. We report those runs separately in Section F.1.

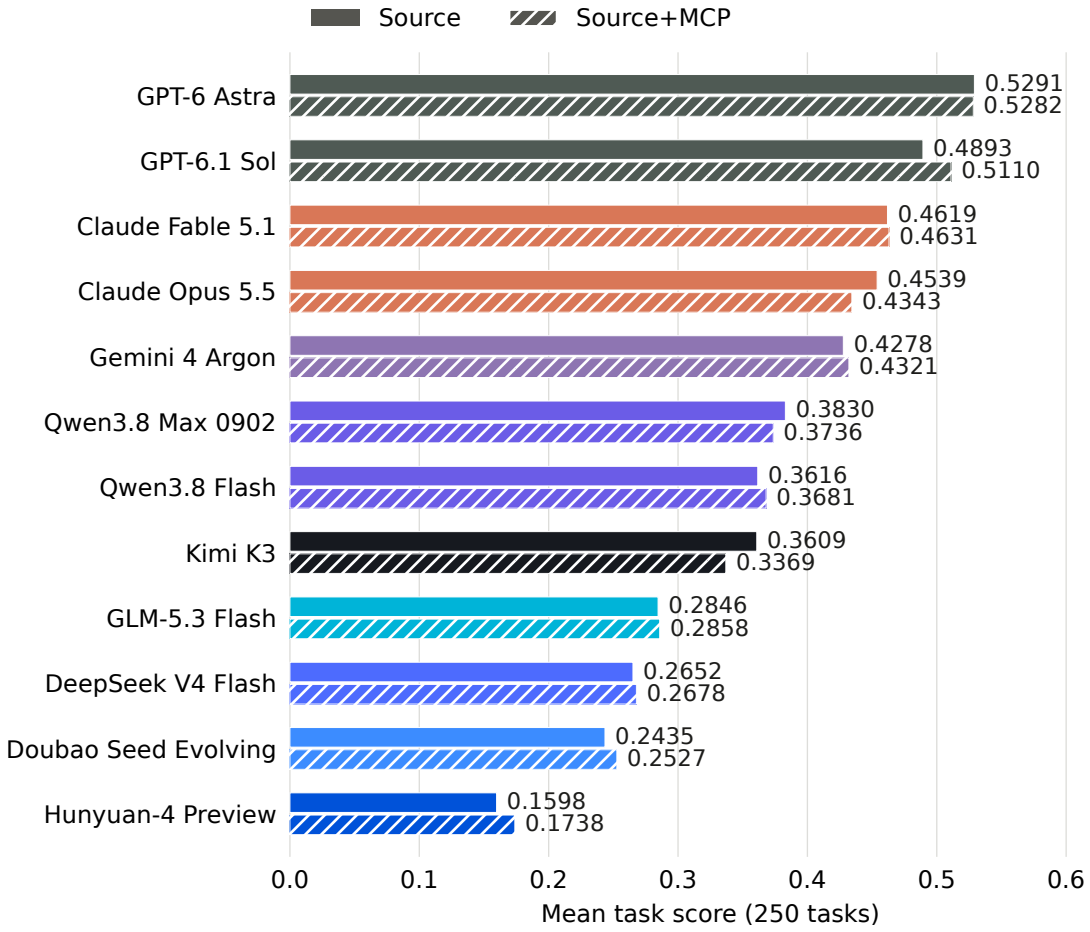


Figure 4: Source and Source+MCP scores for the 12 main model–harness configurations. Solid bars show native source files and code execution; hatched bars add MCP inspection and measurement tools. Scores are displayed to four decimals; colors identify model providers. These configurations do not include the supplementary cross-harness runs. Complete scores for all three settings are in Appendix E.1.

Access settings. Source permits analysis of native 3D data; Source+MCP adds optional inspection and measurement tools. Fixed6 supplies six fixed rendered views. Medical tasks pair renders and slices, giving twelve images; the available records cannot verify the inputs of every retained run.

Execution settings. Table 4 lists the declared harness and known reasoning effort for each model. The assignments cover all three settings, but Fixed6 records do not independently verify them for every run. Claude Code effort is not recorded. Section 3 and Appendix D give repeat counts and comparison limits.

Table 4: Models, declared harnesses, and known source-access reasoning effort. Row order matches Table 1; incomplete execution records limit verification of these settings.

Model	Declared harness	Source-access effort
GPT-6 Astra	Codex	Maximum (reported)
GPT-6.1 Sol	Codex	Maximum (reported)
Claude Fable 5.1	Claude Code	Not recorded
Claude Opus 5.5	Claude Code	Not recorded
Gemini 4 Argon	Claude Code	Not recorded
Qwen3.8 Max 0902	Claude Code	Not recorded
Qwen3.8 Flash	Claude Code	Not recorded
Kimi K3	Claude Code	Not recorded
GLM-5.3 Flash	Claude Code	Not recorded
DeepSeek V4 Flash	Claude Code	Not recorded
Doubao Seed Evolving	Claude Code	Not recorded
Hunyuan-4 Preview	Claude Code	Not recorded

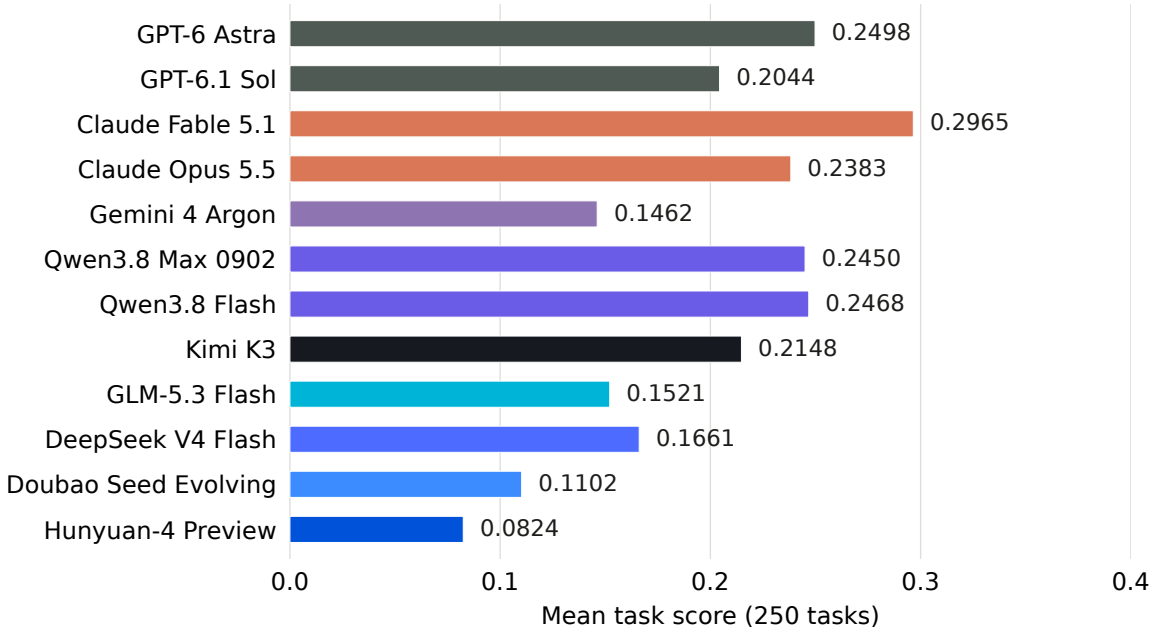


Figure 5: Fixed6 scores for the same 12 configurations, each using six fixed rendered views. Scores match Table 1 and are displayed to four decimals. Source and Source+MCP are shown in Figure 4.

E.2 Condition-specific domain scores

Figure 6 compares Harness + 3D (Source) and Harness + 3D + MCP (Source+MCP) side by side, emphasizing the MCP-associated change for each model and domain while retaining absolute scores. Figure 7 completes the comparison with Fixed6. Together the figures show all 180 means for 12 models, three access conditions and five domains. Each cell averages 50 tasks after averaging the available repeats within each task. Because domains have equal size, their unrounded means recover the overall score. These values describe the selected questions and tested configurations, rather than population-level differences in disciplinary difficulty.

E.2.1 Domain variation

Domain scores show changes that cancel in the overall mean (Figure 6). With MCP, Astra's biology score rises from 0.6200 to 0.6778, while medical imaging falls from 0.3783 to 0.3406. Sol's biology score rises from 0.6000 to 0.6667. Among the canonical configurations, Fable 5.1 with MCP has the highest domain mean, 0.6989 in biology. Appendix E provides the domain scores for all three conditions.

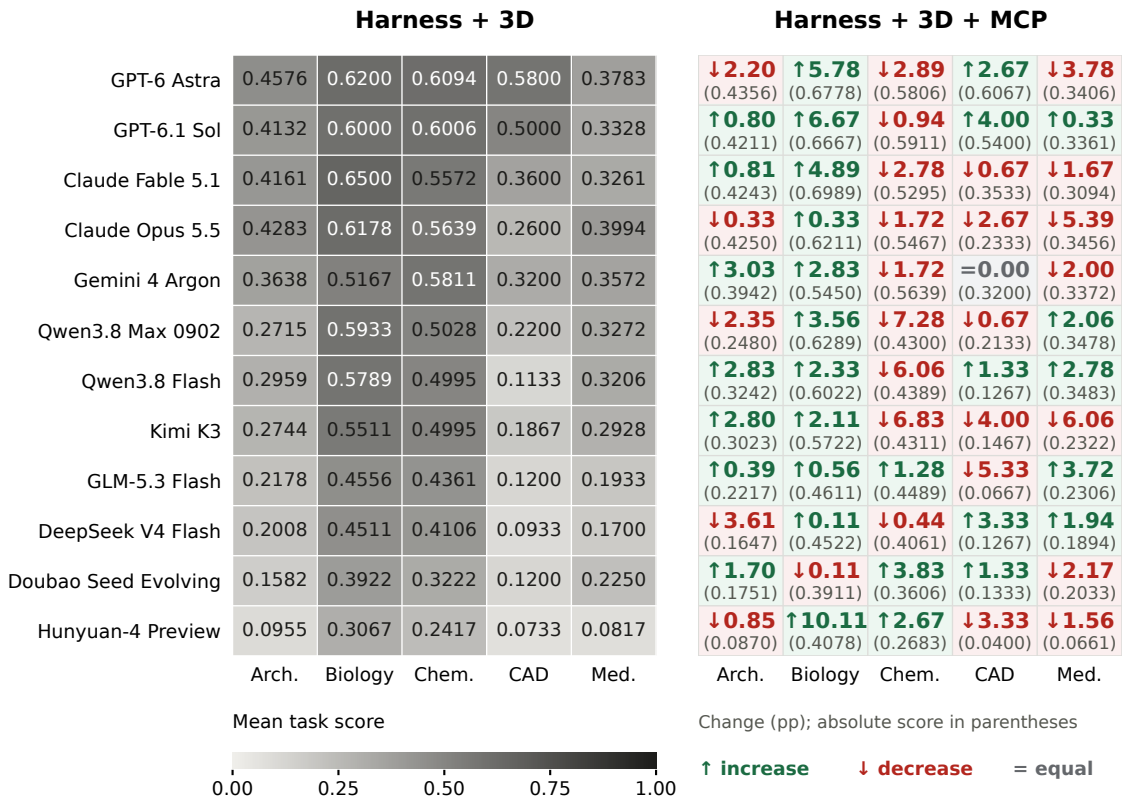


Figure 6: Domain comparison for the 12 model–harness configurations in Table 4. Harness + 3D (Source, left) shows absolute mean scores on a zero-to-one gray scale. Harness + 3D + MCP (Source+MCP, right) emphasizes changes from the same model's Harness + 3D score: green ↑, red ↓ and gray = indicate increases, decreases and ties, with change magnitudes in percentage points (pp). Pale backgrounds encode direction only; parentheses retain absolute scores to four decimals. Changes are computed before rounding. Scores use the 29 October 2026 frozen release, including the 20 October rerun replacements; available repeats are averaged within each task, then 50 tasks are weighted equally per domain. Arch., Chem., CAD and Med. denote architecture, chemistry, mechanical CAD and medical imaging. Fixed6 scores appear in Figure 7.

E.3 Subdomain Results and Aggregation

For these descriptive cross-model comparisons, we average the 12 canonical models equally within each access condition.

Source–Fixed6 gaps depend on the task family. The two largest cross-model gaps occur in dihedral measurement ($N = 9$; 45.06 percentage points) and conformation/occlusion ($N = 21$; 40.68 points). Every model scores higher with Source in both subdomains. Biological spatial arrangement ($N = 14$) has a much smaller mean gap of 2.48 points, and 5 models score higher with Fixed6.

The Source+MCP difference changes sign within every model. Each of the 12 models has both positive and negative Source+MCP minus Source differences across the 22 subdomains. Molecular contact ($N = 14$) has a mean difference of +8.04 percentage points, with 9 models higher. Dihedral measurement ($N = 9$) has a mean difference of -5.25 points, with 8 models lower. A single overall MCP difference conceals these opposing local changes.

Lesion pointing remains a shared weakness. On the 25 lesion-pointing tasks, mean scores are 0.1011, 0.0967 and 0.0344 for Source, Source+MCP and Fixed6, respectively. All 36 model–condition configurations score lower on lesion pointing than on slice comparison ($N = 12$). The medical domain mean mixes this localization weakness with higher slice-comparison performance.

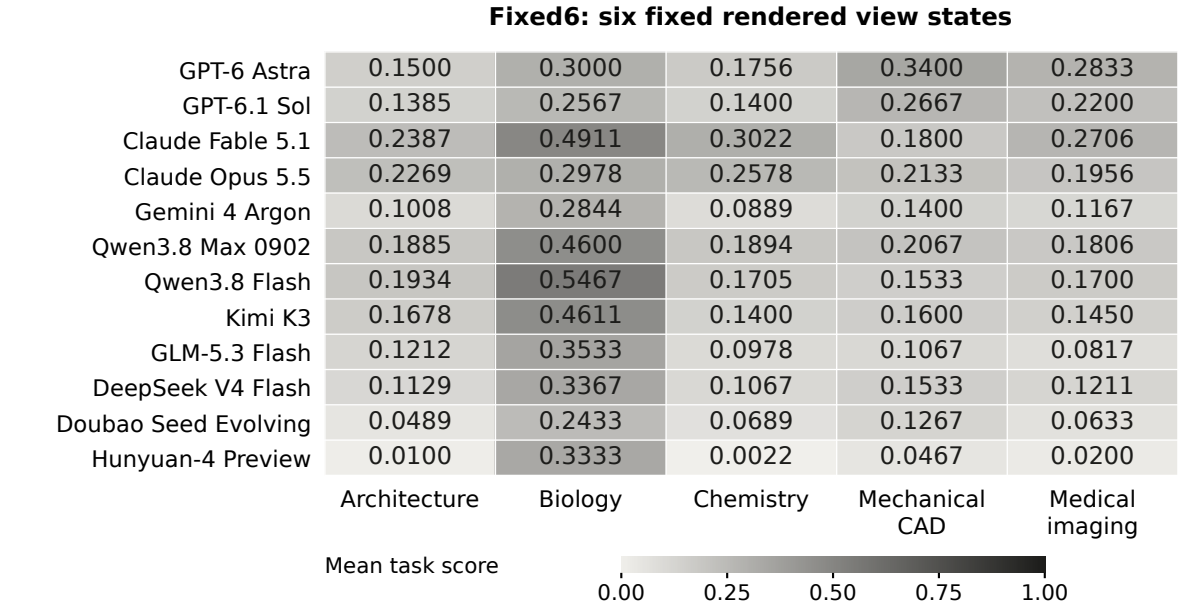


Figure 7: Fixed6 domain mean task scores: six fixed rendered view states, using the same 12 models, five domains, four-decimal display and zero-to-one color scale as Figure 6. Each cell averages 50 equally weighted tasks after averaging the available repeats within each task. Full task-mean precision is retained until display. The score CSV retains all 180 domain means across the three access settings.

Data and aggregation. The six tables give 792 mean scores across 22 subdomains, 12 models, and three access conditions. They use the same 250 tasks and scores as the main comparison. Each task belongs to one subdomain, as recorded in the accompanying metadata. These groups describe task content rather than mutually exclusive reasoning abilities.

We first average available repeats within each task, following Equation 2. For subdomain u with N_u tasks, its mean is $S_u = N_u^{-1} \sum_{i \in u} \bar{s}_i$. Domain and overall means are then

$$S_d = \sum_{u \subset d} \frac{N_u}{50} S_u, \qquad S = \sum_u \frac{N_u}{250} S_u.$$

Subdomains contain 4 to 25 tasks, so weighting by their task counts is necessary to recover the domain and overall means. Calculations retain full precision until the four-decimal display.

Every model–condition pair covers all 250 tasks, and all subdomain cells are populated. Missing repeats are omitted, recorded zero scores are retained, and repeats are averaged within tasks rather than pooled. The resulting totals reproduce all 180 domain means and 36 overall means. Appendix D gives the incomplete-repeat counts and execution-record limits. Small subdomain samples and unmatched execution conditions limit generalization beyond this benchmark and prevent causal attribution of access differences.

Table 5: Source subdomain mean task scores, model group 1 of 2. N is the number of tasks for every model column. Domain and overall rows weight subdomains by N, using unrounded scores. Configurations and repeat handling follow Appendix E.3.

Subdomain	N	GPT-6 Astra	GPT-6.1 Sol	Claude Fable 5.1	Claude Opus 5.5	Gemini 4 Argon	Qwen3.8 Max 0902
Architecture							
Alignment and stacking	12	0.2639	0.2593	0.2569	0.3125	0.2014	0.3056
Dimensional compliance	16	0.4071	0.3551	0.3208	0.3781	0.3521	0.1821
Interference and clearance	12	0.3361	0.3222	0.3685	0.3569	0.3451	0.1801

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Table 5: Source subdomain mean task scores, model group 1 of 2 (continued).

Subdomain	*N*	GPT-6 Astra	GPT-6.1 Sol	Claude Fable 5.1	Claude Opus 5.5	Gemini 4 Argon	Qwen3.8 Max 0902
Object geometry readings	10	0.9167	0.8000	0.8167	0.7333	0.6000	0.4833
Domain mean	50	0.4576	0.4132	0.4161	0.4283	0.3638	0.2715
Biology							
Interaction typing	8	0.8750	0.8750	0.7500	0.7500	0.1250	0.7500
Molecular contact	14	0.7857	0.7143	0.8333	0.8333	0.8214	0.7381
Occluded target geometry	6	0.0000	0.0000	0.2222	0.0926	0.0000	0.2778
Proximity choice	8	0.5000	0.5000	0.5833	0.4583	0.6250	0.5000
Spatial arrangement	14	0.6429	0.6429	0.6310	0.6429	0.5952	0.5476
Domain mean	50	0.6200	0.6000	0.6500	0.6178	0.5167	0.5933
Chemistry							
Conformation occlusion	21	0.6601	0.6283	0.5701	0.5886	0.6905	0.5675
Crystal periodicity	4	0.7222	0.7778	0.7222	0.7222	0.7500	0.6111
Dihedral measurement	9	0.5556	0.5556	0.4630	0.5556	0.5556	0.5370
Hybridization state	7	0.8095	0.7778	0.9048	0.7778	0.5794	0.4921
Molecular extent	9	0.3395	0.3642	0.2778	0.2778	0.2778	0.2778
Domain mean	50	0.6094	0.6006	0.5572	0.5639	0.5811	0.5028
Mechanical CAD							
Assembly composition count	10	0.4333	0.4000	0.3000	0.2000	0.4833	0.1667
Mechanism motion	6	0.5000	0.5000	0.5000	0.4444	0.2778	0.3333
Part instance count	20	0.5667	0.3667	0.1667	0.0500	0.0750	0.0667
Part property identification	14	0.7381	0.7619	0.6190	0.5238	0.5714	0.4286
Domain mean	50	0.5800	0.5000	0.3600	0.2600	0.3200	0.2200
Medical imaging							
Anatomy pointing	5	0.8000	0.8000	0.4667	0.6667	0.6667	0.6667
Geometry measurement	8	0.5938	0.5799	0.4965	0.5382	0.5035	0.5243
Lesion pointing	25	0.2133	0.1467	0.1600	0.1867	0.1333	0.1467
Slice comparison	12	0.4028	0.3611	0.5000	0.6389	0.5972	0.4306
Domain mean	50	0.3783	0.3328	0.3261	0.3994	0.3572	0.3272
Overall	250	**0.5291**	**0.4893**	**0.4619**	**0.4539**	**0.4278**	**0.3830**

Table 6: Source subdomain mean task scores, model group 2 of 2. *N* is the number of tasks for every model column. Domain and overall rows weight subdomains by *N*, using unrounded scores. Configurations and repeat handling follow Appendix E.3.

Subdomain	*N*	Qwen3.8 Flash	Kimi K3	GLM-5.3 Flash	DeepSeek V4 Flash	Doubao Seed Evolving	Hunyuan-4 Preview
Architecture							
Alignment and stacking	12	0.2569	0.2315	0.2116	0.1505	0.1829	0.1319
Dimensional compliance	16	0.2886	0.3179	0.2585	0.2177	0.1836	0.0275
Interference and clearance	12	0.3551	0.3074	0.2125	0.2014	0.1343	0.1319
Object geometry readings	10	0.2833	0.2167	0.1667	0.2333	0.1167	0.1167
Domain mean	50	0.2959	0.2744	0.2178	0.2008	0.1582	0.0955
Biology							
Interaction typing	8	0.7500	0.6250	0.1250	0.1250	0.0833	0.1250
Molecular contact	14	0.7619	0.7143	0.7619	0.7143	0.6905	0.5238
Occluded target geometry	6	0.1296	0.0370	0.1296	0.0926	0.0741	0.0000
Proximity choice	8	0.4583	0.5417	0.4583	0.4167	0.3333	0.3750

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Table 6: Source subdomain mean task scores, model group 2 of 2 (continued).

Subdomain	*N*	Qwen3.8 Flash	Kimi K3	GLM-5.3 Flash	DeepSeek V4 Flash	Doubao Seed Evolving	Hunyuan-4 Preview
Spatial arrangement	14	0.5595	0.5714	0.4762	0.5476	0.4405	0.2857
Domain mean	50	0.5789	0.5511	0.4556	0.4511	0.3922	0.3067
Chemistry							
Conformation occlusion	21	0.5781	0.5437	0.4881	0.4008	0.2275	0.1124
Crystal periodicity	4	0.6667	0.6667	0.4444	0.6111	0.3333	0.2222
Dihedral measurement	9	0.4259	0.5000	0.3889	0.3889	0.3889	0.4074
Hybridization state	7	0.5238	0.5555	0.5397	0.5238	0.7143	0.6667
Molecular extent	9	0.2963	0.2778	0.2778	0.2778	0.1667	0.0556
Domain mean	50	0.4995	0.4995	0.4361	0.4106	0.3222	0.2417
Mechanical CAD							
Assembly composition count	10	0.0667	0.1000	0.0333	0.1000	0.1000	0.0000
Mechanism motion	6	0.2222	0.3889	0.1111	0.1111	0.0000	0.0000
Part instance count	20	0.0500	0.0667	0.1167	0.0500	0.0667	0.0000
Part property identification	14	0.1905	0.3333	0.1905	0.1429	0.2619	0.2619
Domain mean	50	0.1133	0.1867	0.1200	0.0933	0.1200	0.0733
Medical imaging							
Anatomy pointing	5	0.6667	0.5333	0.4667	0.1333	0.2667	0.0000
Geometry measurement	8	0.4410	0.4757	0.3958	0.2500	0.5104	0.1146
Lesion pointing	25	0.0933	0.0400	0.0000	0.0267	0.0400	0.0267
Slice comparison	12	0.5694	0.5972	0.3472	0.4306	0.4028	0.2083
Domain mean	50	0.3206	0.2928	0.1933	0.1700	0.2250	0.0817
Overall	250	**0.3616**	**0.3609**	**0.2846**	**0.2652**	**0.2435**	**0.1598**

Table 7: Source+MCP subdomain mean task scores, model group 1 of 2. *N* is the number of tasks for every model column. Domain and overall rows weight subdomains by *N*, using unrounded scores. Configurations and repeat handling follow Appendix E.3.

Subdomain	*N*	GPT-6 Astra	GPT-6.1 Sol	Claude Fable 5.1	Claude Opus 5.5	Gemini 4 Argon	Qwen3.8 Max 0902
Architecture							
Alignment and stacking	12	0.2708	0.2662	0.2708	0.3125	0.2639	0.2245
Dimensional compliance	16	0.3923	0.3539	0.3081	0.3921	0.3729	0.2025
Interference and clearance	12	0.3130	0.3361	0.3778	0.3384	0.3535	0.2750
Object geometry readings	10	0.8500	0.8167	0.8500	0.7167	0.6333	0.3167
Domain mean	50	0.4356	0.4211	0.4243	0.4250	0.3942	0.2480
Biology							
Interaction typing	8	0.8750	0.8750	0.7500	0.7083	0.1250	0.7500
Molecular contact	14	0.9286	0.9048	0.8810	0.8571	0.7857	0.8810
Occluded target geometry	6	0.0926	0.1111	0.4630	0.0926	0.2778	0.1852
Proximity choice	8	0.5417	0.5000	0.6250	0.5417	0.6250	0.5417
Spatial arrangement	14	0.6429	0.6429	0.6310	0.6071	0.6131	0.5476
Domain mean	50	0.6778	0.6667	0.6989	0.6211	0.5450	0.6289
Chemistry							
Conformation occlusion	21	0.6230	0.6561	0.5886	0.5556	0.6601	0.4841
Crystal periodicity	4	0.8333	0.8056	0.6945	0.6945	0.7084	0.5000
Dihedral measurement	9	0.5000	0.5000	0.4259	0.5000	0.5556	0.4074
Hybridization state	7	0.7460	0.7143	0.7143	0.8413	0.5714	0.4286
Molecular extent	9	0.3210	0.3395	0.2778	0.2778	0.2778	0.2963
Domain mean	50	0.5806	0.5911	0.5295	0.5467	0.5639	0.4300

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Table 7: Source+MCP subdomain mean task scores, model group 1 of 2 (continued).

Subdomain	*N*	GPT-6 Astra	GPT-6.1 Sol	Claude Fable 5.1	Claude Opus 5.5	Gemini 4 Argon	Qwen3.8 Max 0902
Mechanical CAD							
Assembly composition count	10	0.4000	0.4000	0.3000	0.1667	0.4167	0.1000
Mechanism motion	6	0.5000	0.5000	0.4444	0.3333	0.3889	0.3333
Part instance count	20	0.6333	0.4500	0.1500	0.0500	0.0750	0.0833
Part property identification	14	0.7619	0.7857	0.6429	0.5000	0.5714	0.4286
Domain mean	50	0.6067	0.5400	0.3533	0.2333	0.3200	0.2133
Medical imaging							
Anatomy pointing	5	0.8000	0.7333	0.5333	0.6000	0.6667	0.6000
Geometry measurement	8	0.5243	0.4965	0.5382	0.5556	0.4410	0.5903
Lesion pointing	25	0.1600	0.1733	0.1333	0.1467	0.1467	0.1333
Slice comparison	12	0.4028	0.4028	0.4306	0.5139	0.5278	0.5278
Domain mean	50	0.3406	0.3361	0.3094	0.3456	0.3372	0.3478
Overall	250	**0.5282**	**0.5110**	**0.4631**	**0.4343**	**0.4321**	**0.3736**

Table 8: Source+MCP subdomain mean task scores, model group 2 of 2. *N* is the number of tasks for every model column. Domain and overall rows weight subdomains by *N*, using unrounded scores. Configurations and repeat handling follow Appendix E.3.

Subdomain	*N*	Qwen3.8 Flash	Kimi K3	GLM-5.3 Flash	DeepSeek V4 Flash	Doubao Seed Evolving	Hunyuan-4 Preview
Architecture							
Alignment and stacking	12	0.2500	0.2315	0.2245	0.2060	0.2245	0.1134
Dimensional compliance	16	0.3445	0.3455	0.2512	0.1684	0.2175	0.1033
Interference and clearance	12	0.3778	0.3454	0.1977	0.2000	0.1181	0.0556
Object geometry readings	10	0.3167	0.2667	0.2000	0.0667	0.1167	0.0667
Domain mean	50	0.3242	0.3023	0.2217	0.1647	0.1751	0.0870
Biology							
Interaction typing	8	0.7083	0.5417	0.1250	0.1250	0.0833	0.1250
Molecular contact	14	0.8571	0.8333	0.6905	0.7143	0.7619	0.7619
Occluded target geometry	6	0.1296	0.1852	0.2037	0.0741	0.0926	0.1481
Proximity choice	8	0.5000	0.5417	0.4583	0.5000	0.2083	0.4167
Spatial arrangement	14	0.5476	0.5119	0.5357	0.5119	0.4286	0.3214
Domain mean	50	0.6022	0.5722	0.4611	0.4522	0.3911	0.4078
Chemistry							
Conformation occlusion	21	0.4815	0.4709	0.4312	0.3585	0.3585	0.2050
Crystal periodicity	4	0.6667	0.5556	0.6667	0.6667	0.3889	0.1111
Dihedral measurement	9	0.3148	0.3148	0.4630	0.3889	0.2778	0.4444
Hybridization state	7	0.5238	0.5397	0.6508	0.5873	0.6825	0.4762
Molecular extent	9	0.2963	0.3148	0.2222	0.2778	0.1852	0.1481
Domain mean	50	0.4389	0.4311	0.4489	0.4061	0.3606	0.2683
Mechanical CAD							
Assembly composition count	10	0.0667	0.0333	0.0667	0.0667	0.0000	0.0333
Mechanism motion	6	0.2222	0.2222	0.0000	0.0556	0.1111	0.0000
Part instance count	20	0.1000	0.0500	0.0833	0.0833	0.0500	0.0000
Part property identification	14	0.1667	0.3333	0.0714	0.2619	0.3571	0.1190
Domain mean	50	0.1267	0.1467	0.0667	0.1267	0.1333	0.0400
Medical imaging							
Anatomy pointing	5	0.5333	0.3333	0.3333	0.1333	0.4667	0.0000

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Table 8: Source+MCP subdomain mean task scores, model group 2 of 2 (continued).

Subdomain	*N*	Qwen3.8 Flash	Kimi K3	GLM-5.3 Flash	DeepSeek V4 Flash	Doubao Seed Evolving	Hunyuan-4 Preview
Geometry measurement	8	0.4688	0.4306	0.3785	0.1840	0.3750	0.1424
Lesion pointing	25	0.1600	0.0267	0.0400	0.0267	0.0133	0.0000
Slice comparison	12	0.5833	0.4861	0.4861	0.5556	0.3750	0.1806
Domain mean	50	0.3483	0.2322	0.2306	0.1894	0.2033	0.0661
Overall	250	**0.3681**	**0.3369**	**0.2858**	**0.2678**	**0.2527**	**0.1738**

Table 9: Fixed6 subdomain mean task scores, model group 1 of 2. *N* is the number of tasks for every model column. Domain and overall rows weight subdomains by *N*, using unrounded scores. Configurations and repeat handling follow Appendix E.3.

Subdomain	*N*	GPT-6 Astra	GPT-6.1 Sol	Claude Fable 5.1	Claude Opus 5.5	Gemini 4 Argon	Qwen3.8 Max 0902
Architecture							
Alignment and stacking	12	0.1528	0.1361	0.1815	0.1894	0.0972	0.1931
Dimensional compliance	16	0.1251	0.1085	0.2211	0.1851	0.0875	0.1172
Interference and clearance	12	0.1806	0.1713	0.2963	0.3009	0.1644	0.2417
Object geometry readings	10	0.1500	0.1500	0.2667	0.2500	0.0500	0.2333
Domain mean	50	0.1500	0.1385	0.2387	0.2269	0.1008	0.1885
Biology							
Interaction typing	8	0.3750	0.2917	0.7500	0.3750	0.1250	0.6667
Molecular contact	14	0.0000	0.0000	0.4524	0.0000	0.0952	0.5000
Occluded target geometry	6	0.0000	0.0000	0.3148	0.0926	0.1481	0.1667
Proximity choice	8	0.4583	0.4167	0.2500	0.4167	0.5833	0.4583
Spatial arrangement	14	0.5952	0.5119	0.5952	0.5714	0.4524	0.4286
Domain mean	50	0.3000	0.2567	0.4911	0.2978	0.2844	0.4600
Chemistry							
Conformation occlusion	21	0.1058	0.0794	0.2883	0.2010	0.0529	0.1468
Crystal periodicity	4	0.0000	0.0000	0.5000	0.5000	0.3889	0.4722
Dihedral measurement	9	0.0741	0.0741	0.0370	0.0000	0.0000	0.0370
Hybridization state	7	0.8413	0.6667	0.7460	0.8095	0.2540	0.5238
Molecular extent	9	0.0000	0.0000	0.1667	0.1111	0.0000	0.0556
Domain mean	50	0.1756	0.1400	0.3022	0.2578	0.0889	0.1894
Mechanical CAD							
Assembly composition count	10	0.4333	0.3000	0.3000	0.2667	0.1333	0.3333
Mechanism motion	6	0.4444	0.3333	0.5000	0.3333	0.2778	0.2778
Part instance count	20	0.0500	0.0500	0.0000	0.0000	0.0000	0.0000
Part property identification	14	0.6429	0.5238	0.2143	0.4286	0.2857	0.3810
Domain mean	50	0.3400	0.2667	0.1800	0.2133	0.1400	0.2067
Medical imaging							
Anatomy pointing	5	0.6667	0.7333	0.7333	0.5333	0.0000	0.1333
Geometry measurement	8	0.0417	0.0417	0.2535	0.1389	0.1250	0.2326
Lesion pointing	25	0.2000	0.1200	0.0533	0.0267	0.0000	0.0000
Slice comparison	12	0.4583	0.3333	0.5417	0.4444	0.4028	0.5417
Domain mean	50	0.2833	0.2200	0.2706	0.1956	0.1167	0.1806
Overall	250	**0.2498**	**0.2044**	**0.2965**	**0.2383**	**0.1462**	**0.2450**

Table 10: Fixed6 subdomain mean task scores, model group 2 of 2. N is the number of tasks for every model column. Domain and overall rows weight subdomains by N, using unrounded scores. Configurations and repeat handling follow Appendix E.3.

Subdomain	N	Qwen3.8 Flash	Kimi K3	GLM-5.3 Flash	DeepSeek V4 Flash	Doubao Seed Evolving	Hunyuan-4 Preview
Architecture							
Alignment and stacking	12	0.1486	0.1259	0.0810	0.1343	0.0694	0.0139
Dimensional compliance	16	0.1680	0.1600	0.0568	0.0925	0.0104	0.0000
Interference and clearance	12	0.3083	0.2625	0.1954	0.1574	0.1065	0.0278
Object geometry readings	10	0.1500	0.1167	0.1833	0.0667	0.0167	0.0000
Domain mean	50	0.1934	0.1678	0.1212	0.1129	0.0489	0.0100
Biology							
Interaction typing	8	0.7083	0.3750	0.0417	0.0833	0.0000	0.0833
Molecular contact	14	0.6429	0.5952	0.5000	0.4524	0.3571	0.5714
Occluded target geometry	6	0.2222	0.0926	0.0556	0.0000	0.0000	0.0000
Proximity choice	8	0.4167	0.4583	0.2500	0.2500	0.0833	0.2500
Spatial arrangement	14	0.5714	0.5357	0.5714	0.5595	0.4643	0.4286
Domain mean	50	0.5467	0.4611	0.3533	0.3367	0.2433	0.3333
Chemistry							
Conformation occlusion	21	0.1468	0.0529	0.0370	0.0635	0.0000	0.0000
Crystal periodicity	4	0.4166	0.4167	0.1944	0.1667	0.0556	0.0000
Dihedral measurement	9	0.0185	0.0741	0.0000	0.0000	0.0000	0.0000
Hybridization state	7	0.4921	0.5079	0.4762	0.4762	0.4603	0.0159
Molecular extent	9	0.0185	0.0000	0.0000	0.0000	0.0000	0.0000
Domain mean	50	0.1705	0.1400	0.0978	0.1067	0.0689	0.0022
Mechanical CAD							
Assembly composition count	10	0.3000	0.1333	0.1667	0.3667	0.1333	0.0000
Mechanism motion	6	0.2778	0.4444	0.0000	0.1111	0.1111	0.0000
Part instance count	20	0.0000	0.0000	0.0000	0.0167	0.0000	0.0000
Part property identification	14	0.2143	0.2857	0.2619	0.2143	0.3095	0.1667
Domain mean	50	0.1533	0.1600	0.1067	0.1533	0.1267	0.0467
Medical imaging							
Anatomy pointing	5	0.1333	0.0000	0.0000	0.0000	0.0000	0.0000
Geometry measurement	8	0.1458	0.1562	0.0521	0.0694	0.0000	0.0000
Lesion pointing	25	0.0000	0.0133	0.0000	0.0000	0.0000	0.0000
Slice comparison	12	0.5556	0.4722	0.3056	0.4583	0.2639	0.0833
Domain mean	50	0.1700	0.1450	0.0817	0.1211	0.0633	0.0200
Overall	250	**0.2468**	**0.2148**	**0.1521**	**0.1661**	**0.1102**	**0.0824**

E.4 Task score profile

For each task, we take the highest mean across the 36 main configurations. Table 11 groups tasks by whether that maximum is zero, between zero and one, or one. The counts are 34, 83, and 133. A maximum of one requires a full-score mean for at least one configuration; a single successful repeat is insufficient.

Across the 36 configurations, 34 tasks receive a mean score of zero everywhere. At least one configuration reaches a task mean of one on 133 tasks. The remaining 83 tasks have a best mean between zero and one. Medical imaging accounts for 13 all-zero tasks, compared with four to six in each other domain (Figure 8). These counts characterize the selected tasks and tested systems, rather than intrinsic domain difficulty.

Reference inspections show how an expert approached a question, but cannot establish what an eval-

Table 11: Descriptive score profile across the 36 canonical configurations. Each task is assigned according to its highest available-repeat mean: zero, strictly between zero and one, or one. A full mean for one configuration does not imply success for every model or individual repeat.

Domain	All zero	Partial maximum	Full-score maximum
Architecture	6	27	17
Biology	4	11	35
Chemistry	6	21	23
Mechanical CAD	5	10	35
Medical imaging	13	14	23

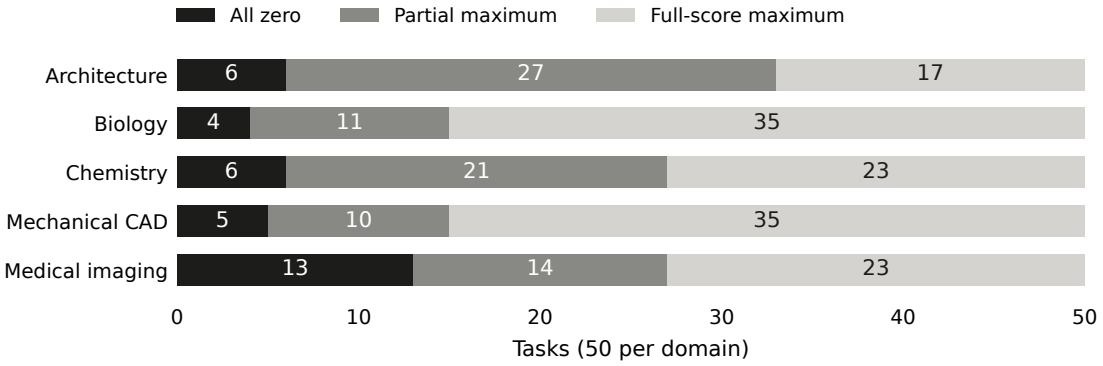


Figure 8: Task score profiles within each domain. Every task is assigned by its highest available-repeat mean across the 36 canonical configurations: zero (dark), strictly between zero and one (mid-grey), or one (light). Segment labels count tasks, with 50 tasks per row. These are descriptive categories for the evaluated cohort. Table 11 gives the counts.

uated model observed. Answer scores alone also cannot distinguish failures to acquire evidence from failures of scientific inference.

E.5 CAD target-definition sensitivity

The prior audit examined eight CAD questions with unclear targets or counting units. Seven stored counts matched the number of solid bodies across the assembled object. For the remaining question, the reference of seven counted distinct immediate component definitions, whereas the whole assembly contained 26 solid bodies. Without an explicit target and counting unit, reproducing a stored number does not resolve the intended answer. The audit therefore did not establish a numerical reference error.

The diagnostic excludes these eight questions and weights the remaining 242 tasks equally: 42 CAD tasks and 50 in each other domain. It preserves all 12-model rankings, every Source–Fixed6 direction, and all MCP difference signs. The largest mean change is 1.531 percentage points for Fable Source+MCP. The benchmark questions, references, and reported 250-task scores remain unchanged.

Supporting materials identify the eight questions and document the independent component counts. Recomputing this score diagnostic uses the recorded scores; repeating the structural audit additionally requires the original CAD objects.

E.6 Scope of the execution-configuration comparison

The harness comparison in Section F.1 averages three repeats per task and weights all 250 tasks equally. We report the observed differences without a paired significance test or a causal estimate of harness effects.

E.7 Additional protocol comparison details

Source and Fixed6 differ in both evidence representation and permitted computation. Run records also leave uncertainty about whether execution conditions, inputs, and scoring were fully matched. We

Table 12: Published 250-task means and the 242-task diagnostic excluding eight CAD questions with underspecified targets or counting units. The diagnostic weights 42 CAD and 50 tasks from each other domain equally per task. The exclusion neither changes the published benchmark nor certifies the remaining references.

	Source		Source+MCP		Fixed6	
Model	250	242	250	242	250	242
GPT-6 Astra	0.5291	0.5356	0.5282	0.5292	0.2498	0.2580
GPT-6.1 Sol	0.4893	0.5013	0.5110	0.5183	0.2044	0.2097
Claude Fable 5.1	0.4619	0.4772	0.4631	0.4784	0.2965	0.3063
Claude Opus 5.5	0.4539	0.4689	0.4343	0.4487	0.2383	0.2461
Gemini 4 Argon	0.4278	0.4419	0.4321	0.4463	0.1462	0.1510
Qwen3.8 Max 0902	0.3830	0.3956	0.3736	0.3860	0.2450	0.2531
Qwen3.8 Flash	0.3616	0.3736	0.3681	0.3802	0.2468	0.2550
Kimi K3	0.3609	0.3728	0.3369	0.3481	0.2148	0.2219
GLM-5.3 Flash	0.2846	0.2940	0.2858	0.2952	0.1521	0.1572
DeepSeek V4 Flash	0.2652	0.2739	0.2678	0.2767	0.1661	0.1703
Doubao Seed Evolving	0.2435	0.2516	0.2527	0.2610	0.1102	0.1139
Hunyuan-4 Preview	0.1598	0.1650	0.1738	0.1796	0.0824	0.0852

therefore interpret score differences as comparisons of evaluated configurations. They do not isolate effects of active observation, MCP access, or the model alone.

E.7.1 Additional score comparisons

A prior audit of eight CAD questions found underspecified targets or counting units, without establishing numerical errors in their reference answers. Excluding those questions leaves 242 tasks and preserves all three model rankings, every Source–Fixed6 direction, and the eight-increase/four-decrease MCP split.

Astra's Source score changes from 0.5291 to 0.5356; Fable's Fixed6 score changes from 0.2965 to 0.3063. Appendix E gives the complete diagnostic. This check leaves the benchmark unchanged and does not validate all remaining references.

F Execution Configurations and Earlier Design Studies

We examine execution settings and two earlier studies of view budgets and tool-only access. The earlier studies used different task sets and cannot estimate matched effects on the present benchmark.

F.1 Execution harness and reasoning budget

The alternative-harness runs cover Source and Source+MCP for four models (Table 13). GPT models use maximum reasoning effort in Codex. The supplementary Fable and Opus runs use high effort because maximum-effort requests encountered timeouts. The main leaderboard retains one declared harness per model, with these supplementary runs reported separately.

Codex scores higher in all eight model–access comparisons, by 4.56–13.40 percentage points. Supplementary Opus/Codex Source reaches 0.5448, exceeding the main Source maximum of 0.5291. Differences in execution settings and evaluation records prevent attribution to a controlled harness effect. Appendix D.2 describes the comparison data.

Additional interaction sometimes helps an agent recover from an error. In one Fable/Codex trial, a failed attempt to read molecular data led to further inspection and an alternative distance calculation. The answer received full credit under the recorded rubric. Yet among 3,000 supplementary Fable/Opus Codex trials, full-score trials had a median of six model responses, compared with nine for zero-score trials. In that batch, 47 trials ended in timeouts. These associations do not establish a benefit from longer interaction, and response length does not measure hidden reasoning.

The systems also organize inspection and computation differently. Codex can combine these operations, whereas Claude Code exposes separate operations for reading, editing, and execution. Available records

do not establish whether these differences explain the scores; that would require matched trajectories from both systems. Appendix D.3 gives the recovery example and counting definitions.

Table 13: Cross-harness performance on 250 tasks.

Model	Access	Codex	Claude Code	Δ (pp)
GPT-6 Astra	Source	0.5291	0.4082	+12.09
	Source+MCP	0.5282	0.3942	+13.40
GPT-6.1 Sol	Source	0.4893	0.3865	+10.28
	Source+MCP	0.5110	0.3843	+12.67
Claude Fable 5.1	Source	0.5075	0.4619	+4.56
	Source+MCP	0.5349	0.4631	+7.18
Claude Opus 5.5	Source	0.5448	0.4539	+9.10
	Source+MCP	0.5322	0.4343	+9.78

Δ denotes Codex − Claude Code in percentage points.

F.2 Fixed-view budget and camera plan

In the earlier camera study, more views did not consistently improve scores. Aligned views changed little from three to six (0.1516 to 0.1529) and decreased at twelve; orbit views tied at six and twelve (Table 14). Six views provide a practical comparison point for that study, but are not an established optimum for the present benchmark.

The study tested GPT-6 Astra with Claude Code and LiteLLM on 250 tasks, using maximum reasoning effort and retaining the last attempt. It compared aligned and orbit views at $N \in \{1, 3, 6, 12\}$. The present evaluation includes only six fixed views. The task set differs, and matching inputs and scoring conditions are not established. We therefore report the earlier pattern separately. Appendix F provides the full results and the medical image-count exception.

Table 14: Mean task scores in the earlier camera-plan study using GPT-6 Astra with Claude Code and LiteLLM.

Camera plan	$N = 1$	$N = 3$	$N = 6$	$N = 12$
Aligned	0.1223	0.1516	0.1529	0.1427
Orbit	0.1171	0.1204	0.1463	0.1463

Each cell averages 250 task scores at maximum reasoning effort, retaining the last attempt. N is the nominal number of views; medical inputs contain $2N$ images. The present evaluation includes only six fixed views. Appendix F gives full diagnostics and comparison limits.

F.3 Tool-only access versus source computation

Source+MCP permits both source computation and viewer inspection. An MCP-only condition instead restricts evidence acquisition to the inspection interface. The earlier study included this condition, giving a mean score of 0.1757 for GPT-6 Astra across 250 tasks (Table 15).

That run used Claude Code and LiteLLM at maximum reasoning effort and retained the last attempt. Although it was described as tool-only, the available records cannot verify that file access and direct computation were fully disabled. Its task set and execution conditions also differ from the present evaluation. The score therefore illustrates a distinct access protocol, without estimating the benefit of MCP in Source+MCP. Appendix F gives the remaining limits.

Table 15: Earlier MCP-only and main source-access scores for GPT-6 Astra. The access protocols, task sets, and execution conditions are not matched, so these scores cannot estimate the effect of adding MCP.

Cohort ($N = 250$)	Access	Score
Earlier study	MCP-only	0.1757
Main evaluation	Source	0.5291
	Source+MCP	0.5282

Earlier and main results use different cohorts and harnesses.

F.4 Camera-plan comparison

The camera-plan comparison above gives the main pattern and mean scores. Table 16 adds per-condition diagnostics and separate present-day Fixed6 scores. The earlier study contains 250 tasks under eight camera conditions and one MCP-only condition. It shares 205 task identifiers with the present benchmark; the 45 replacement questions are absent.

The present evaluation includes only six fixed views and provides no matched camera-plan comparison. The earlier documentation reports $2N$ medical images per task, half identical between aligned and orbit plans. This limits interpretation of medical camera differences. The original rendering records are unavailable for independent verification.

Table 16: Earlier camera-plan results and present Fixed6 scores. Panel A compares eight camera conditions; Panel B gives selected Fixed6 scores from the main evaluation. Task sets, execution conditions, and repeat policies are not matched, so differences between panels cannot estimate a camera effect.

Panel A: earlier study, GPT-6 Astra with Claude Code and LiteLLM

Plan	N	Mean	Full	Zero	Ineligible	Retried
Aligned	1	0.1223	24	210	8	10
Aligned	3	0.1516	29	198	8	9
Aligned	6	0.1529	29	198	8	6
Aligned	12	0.1427	26	199	8	2
Orbit	1	0.1171	23	211	8	3
Orbit	3	0.1204	22	207	8	5
Orbit	6	0.1463	27	199	8	4
Orbit	12	0.1463	28	199	8	2

Panel B: selected Fixed6 scores from the main evaluation

Model	Mean task score
GPT-6 Astra	0.2498
GPT-6.1 Sol	0.2044
Claude Fable 5.1	0.2965
Claude Opus 5.5	0.2383

Panel A: Each condition contains 250 tasks, uses maximum reasoning effort, and retains the last attempt. The earlier task set shares 205 identifiers with the present benchmark; the 45 replacement tasks are absent. Questions, references, scoring, and camera inputs are not fully matched. Full and Zero count retained scores of one and zero. Ineligible and Retried count tasks marked ineligible and tasks retried in the original records. These counts can overlap Zero and do not change the 250-task denominator. Medical tasks contain $2N$ images, reportedly with half identical between camera plans; original rendering records are unavailable for verification. **Panel B:** The main evaluation covers 12 models and 250 tasks. Eleven models have three scores per task. Hunyuan has two scores for three tasks and three for the other 247. Fixed6 records do not verify the harness for every run. The main Astra configuration uses Codex, while the earlier study used Claude Code and LiteLLM. Only six views were tested in the main evaluation, so these results do not establish an optimal view budget.

F.5 Tool-only access

Table 15 and the tool-only comparison above describe the earlier MCP-only condition, with a mean of 0.1756667 across 250 tasks. Its documentation states that agents could inspect raw objects only through the MCP interface. However, available tool restrictions and trajectories are insufficient to verify that direct file access and computation were disabled. This differs from Source+MCP, which explicitly permits both source computation and inspection. No matched comparison with the present task set is available.